

```

#include<stdio.h>
#include<stdlib.h>
#include<string.h>

#define ALPHABET_SIZE 26

struct TrieNode{
    struct TrieNode * children[ALPHABET_SIZE];
    int isEndOfWord;
};

struct TrieNode *createNode(){
    struct TrieNode *node = (struct TrieNode *) malloc(sizeof(struct TrieNode));

    node->isEndOfWord = 0;

    for(int i =0; i < ALPHABET_SIZE; i++){
        node->children[i] = NULL;
    }

    return node;
};

void insert(struct TrieNode *root, const char *word){
    struct TrieNode *current = root;
    while(*word){
        int index = *word - 'a';
        if(!current->children[index]){
            current->children[index] = createNode();
            current = current->children[index];
            word++;
        }

        current->isEndOfWord= 1;
    }
}

```

```

int countChildren(struct TrieNode *node, int *index){
    int count = 0;
    for(int i = 0; i < ALPHABET_SIZE; i++){
        if(node->children[i]){
            count++;
            *index = i;
        }
    }
    return count;
}

void longestCommonPrefix(struct TrieNode *root, char *prefix){
    struct TrieNode * current = root;
    int index;
    int length = 0;

    while(countChildren(current, &index) == 1 && !current->isEndOfWord){
        prefix[length++] = 'a' + index;
        current = current->children[index];
    }
    prefix[length] = '\0';
}

```

```
int main() {
    char *words[] = { "flower", "flow", "flight" };
    int n = sizeof(words) / sizeof(words[0]);

    struct TrieNode *root = createNode();

    for(int i = 0; i < n; i++)
        insert(root, words[i]);

    char prefix[100];
    longestCommonPrefix(root, prefix);

    if(strlen(prefix) > 0) {
        printf("Longest Common Prefix: %s\n", prefix);
    }
    else {
        printf("No common Prefix\n");
    }
}
```

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Longest Common Prefix: fl

Process returned 0 (0x0) execution time : 0.010 s

Press any key to continue.